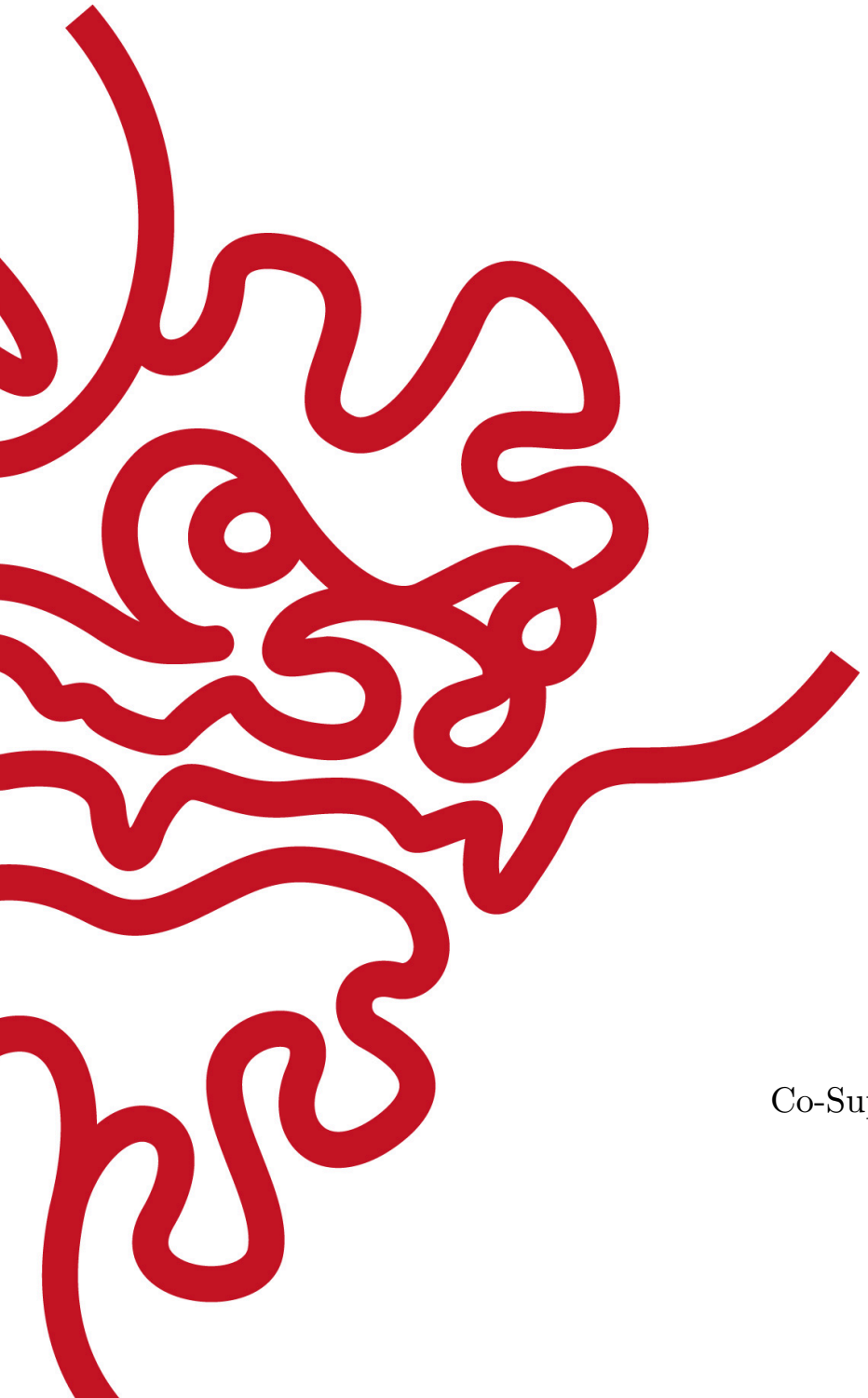
Okinawa Institute of Science And Technology
Graduate University

Thesis submitted for the degree

Doctor of Philosophy

Typst Dissertation Template



by

Kazuma Takada

Supervisor: **S. Upervisor**
Co-Supervisor: **C. O'Supervisor**

September 2025

Declaration of Original and Sole Authorship

I, Kazuma Takada, declare that this thesis entitled *Typst Dissertation Template* and the data presented in it are original and my own work.
I confirm that:

- No part of this work has previously been submitted for a degree at this or any other university.
- References to the work of others have been clearly acknowledged. Quotations from the work of others have been clearly indicated and attributed to them.
- In cases where others have contributed to part of this work, such contribution has been clearly acknowledged and distinguished from my own work.
- None of this work has been previously published or pre-printed elsewhere, with the exception of the following:

  – **Author, A.**, Coauthor, B., & Supervisor, C. (2025). Title of the first journal article that overlaps with this thesis. *Journal Name*, *12*(3), 123456. https://doi.org/10.0000/example.2025.123456
  Contribution: *Conceptualization, Investigation, Formal analysis, Writing – original draft*
  – **Author, A.**, & Supervisor, C. (2025). Title of the second journal article. *Another Journal*, *7*(1), 42. https://doi.org/10.0000/example.2025.000042
  Contribution: *Methodology, Investigation, Formal analysis, Writing – review & editing*

Date: September 2025

Signature: 

Declaration on the Use of AI Tools

I acknowledge the use of generative artificial intelligence (AI) tools in the conduct of my research. Details are as follows:

- **Tools used:** Name of tool and version, Vendor, Inc.
- **Purpose of use:** language editing/proofreading, optimizing and debugging scripts of experiment and analysis

All outputs generated by the AI tools were reviewed, verified, and edited by the author as necessary. These tools were not used to fabricate data, generate research results, or create original scientific ideas presented in this manuscript.

Abstract

The abstract should fit within a page. It should be written in a way that is accessible to a general audience, summarizing the main findings and significance of the research.

Acknowledgments

Please refer to the https://groups.oist.jp/grad/academic-program-policies for specifications.

List of Abbreviations

Please refer to the https://groups.oist.jp/grad/academic-program-policies for specifications.

Here is an example of how to write a list of abbreviations using a bulleted list:

- **OIST**: Okinawa Institute of Science and Technology
- **e.g.**: For example
- **etc.**: And so on
- **PPT**: positive partial transpose
- **SRPT**: Schrödinger-Robertson partial transpose

Glossary

Please refer to the https://groups.oist.jp/grad/academic-program-policies for specifications.

Here is an example of how to write a glossary using a bulleted list:

- **Dipole Blockade**: Phenomenon in which the simultaneous excitation of two atoms is inhibited by their dipolar interaction.
- **Cavity Induced Transparency**: Phenomenon in which a cavity containing two atoms excited with light at a frequency halfway between the atomic frequencies contains the number of photons an empty cavity would contain.

Nomenclature

Please refer to the https://groups.oist.jp/grad/academic-program-policies for specifications.

Here is an example of how to write a list of nomenclature using a bulleted list:

- c: Speed of light ($2.99792458 \times 10^8 \mathrm{ms}^{-1}$)
- $h.|$: Reduced Planck constant ($1.054571817 \times 10^{-34}$ Js)
- k_B: Boltzmann constant ($1.380658 \times 10^{-23} \mathrm{JK}^{-1}$)
- Z_0: Impedance of free space (376.730313461Ω)
- μ_0: Permeability of free-space ($4\pi \times 10^{-7} \mathrm{Hm}^{-1}$)

List of Publications

Publications

Journal (peer-reviewed)

- **Author, A.**, Coauthor, B., & Supervisor, C. (2025). Title of the first journal article that overlaps with this thesis. *Journal Name*, *12*(3), 123456. https://doi.org/10.0000/example.2025.123456

  Contribution: *Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization*

- **Author, A.**, & Supervisor, C. (2025). Title of the second journal article that overlaps with this thesis. *Another Journal*, *7*(1), 42. https://doi.org/10.0000/example.2025.000042

  Contribution: *Writing – review & editing, Methodology, Investigation, Formal analysis*

Presentations

Conference

- Oral **Author, A.**, & Supervisor, C. (2024). Title of the conference talk. *Name of the Conference*, City, Country.

  Contribution: *Presentation, Data curation, Formal analysis*

- Poster **Author, A.**, Coauthor, B., & Supervisor, C. (2023). Title of the poster. *Name of the Meeting*, City, Country.

  Contribution: *Presentation, Investigation, Formal analysis*

Dedication

If desired, an optional and short dedication may be included here.

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Chapter 1

Introduction

This is the introduction. You might want to leave it unnumbered, as it is now. If you want to number it, treat it like any other chapter.

Chapter 2

Guidelines on the Preparation of Thesis

Please refer to https://groups.oist.jp/grad/academic-program-policies for specifications.

Many of the formatting requirements such as page size, fonts, etc are built-in into this template. Do not modify them.

Chapter 3

How to use the templates

This is a practical guide into how to use this template, by explaining the role of the different folders, and an option of `thesis(...mode)` property, which accepts either `draft` or `submission`.

3.1 Folders

The main folder contains three folders detailed here:

- **images.** This folder should contain all the images that you will use in your thesis. It can contain subfolders, for example one for each chapter. To include an image from the main text, you can use the command `#figure(image("path/to/image.jpg", width: 30%))`. The path is relative to the main folder, so if you want to include an image from the images folder, you can write:

```
#figure(
  image("../images/chap3/emblem.jpg", width: 30%),
  caption: [Full caption with all the details here.font size has to
be 10pt, and sentences has to be left side.],
) <fig-example>
```

- **chapters.** This folder contains a series of `LaTeX` files that form the main text: introduction, chapters, conclusion, appendices and published articles. The introduction and conclusion as they are now are not numbered, which creates a few difficulties with the headers of the thesis. If you want the introduction and conclusion to be numbered, re-write and treat them as regular chapters. These text files are included in the main text of the thesis with the command `#include "chapters/filename.typ"` in the `./main.typ`.

```
// main.typ
// Main chapters
#include "chapters/introduction.typ"
#include "chapters/chapter1.typ"
#include "chapters/chapter2.typ"
#include "chapters/chapter3.typ"
#include "chapters/conclusion.typ"
```

- **preamble.** This folder contains a series of typst files with the pages that will appear before the main text. Please write (or copy and paste) your own text in those files and delete the dummy text when appropriate. The files are:
  - `abbreviations.typ` — List of abbreviations. If the list goes over one page, continue it as a new list on the next page.
  - `abstract.typ` — Abstract. Follow directions in the file.
  - `acknowledgments.typ` — Acknowledgments. Follow directions in the file.
  - `declaration.typ` — Declaration of Original and Sole Authorship. Only modify the last item and the signature. See Section 3.3.
  - `declaration_ai_tools.typ` — Declaration on the Use of AI Tools. State which generative AI tools you used and for what. Delete the file and its `#include` line if you used none.
  - `dedication.typ` — Dedication (optional). Should only be a very few lines.
  - `glossary.typ` — Glossary (optional). If the list goes over one page, continue it as a new list on the next page.
  - `nomenclature.typ` — Nomenclature (optional). If the list goes over one page, continue it as a new list on the next page.
  - `publications.typ` — List of Publications and Presentations (optional). Each entry can carry a `#contrib[...]` line stating your own contribution, and conference entries can be tagged with `#entry_type("Oral")` or `#entry_type("Poster")`. All spacing knobs are defined at the top of the file.

3.2 main.typ file

This is **the main file**, the only one that needs to be compiled to build the thesis. The thesis function is executed at the top of this file. When submitting to a graduate school, compile with the `mode` parameter set to `"draft"`.

```
#show: doc => thesis(
  title: "Typst Dissertation Template",
  author: "Kazuma takada",
  supervisor: "S. Upervisor",
  cosupervisor: "C. O'Supervisor",
  submission_date: "August 2025",
  mode: "draft",
  colored_refs: false,
  doc,
)
```

When you submit a temporary version to the graduate school, compile with `"submission"` parameter.

`colored_refs` controls the colour of citations and cross-references. It is `false` (black) by default, which is what you want for the printed and submitted versions. Set it to `true` while writing to get blue, clickable citations and cross-references, which makes it much easier to navigate a long draft on screen.

3.3 Declaration page and signature

The declaration page pulls the title, author, and submission date straight from `thesis(...)`, so you never retype them. Two things do need editing:

- The last confirmation item lists every publication or preprint that overlaps with the thesis, together with a `Contribution:` line for each. If none of your work has appeared elsewhere, delete the nested list and shorten the sentence. If a co-author's work appears in the thesis, obtain a signed release from every affected co-author, attach it to both submitted versions as a separate document, and state here that authorization has been obtained.

- The signature. The template ships a placeholder at `images/signature.svg`. Replace it with a scan of your own signature — any format Typst reads (SVG, PNG, JPG, PDF) works — and adjust `height` in `preamble/declaration.typ` if needed. Note that the whole block has to fit under the publication list, so a taller signature may push it onto a second page.

3.3.1 Page size, Margins, Spacing, Justification, and Pagination

These settings are all defined in `./oist_thesis.typ`. They should not be changed.

> ***Page size**: Theses are to be formatted to A4 page size.*
> ***Margins**: The left margin should be 3 cm. Top, right, and bottom margins should be 2.5 cm.*

> ***Spacing**: Spacing should be 1.5 line spaced for examination and single spacing for final submission.*
> ***Justification**: The main body of text left-right justified. Titles should be left-justified. Equation numbers should be right justified.*
> ***Pagination**: Preliminary pages, ending before the main body of text, may be numbered sequentially using roman numerals. The main body of text, beginning with the first page of the introduction, must be numbered sequentially, including figure pages and blank pages, starting at page number 1 for the first page of the introduction. Arabic numerals are to be used.*
>
> — OIST, Graduate School Policies[1]

3.3.2 Table of Contents, List of Figures, List of Tables

These settings are for displaying the outline of the chapters, tables, and figures that will appear in the paper. Basically, there is no need to change them.

3.3.3 Title Page

The title page is defined by the `#let title_page()` function in `./oist_thesis.typ`. Basically, there is no need to change it.

3.3.4 Preamble Pages

The page numbering settings are specified at the end of the `thesis()` function in `./oist_thesis.typ`. No changes are necessary.

3.3.5 Thesis Main Text

The main text is **Times New Roman 12pt**. This definition is specified as `set text(font: "Times New Roman", lang: "en", size: 12pt)` in the `thesis` function of `./oist_thesis.typ`. This is strictly defined in OIST policy as following.

> ***Font**: Times New Roman, 12-point font should be used for all main body text. For graph legends, titles, image annotations, etc., Arial, Helvetica or Calibri, 10-point font, should be used for presentation clarity. For headings, any font or size may be used for presentation and design considerations.*
> …
> ***Spelling**: American spelling.*
>
> — OIST, Graduate School Policies[2]

[1] https://www.oist.jp/education/policies-regulations/gs-policies
[2] https://www.oist.jp/education/policies-regulations/gs-policies

3.3.6 Bibliography

References should be listed in bibliography.bib. Paste them in bibtex format. Please see the example below. When referencing literature in typst, write it as shown in `@Lee98`. This template uses APA style, but if you want to use another style such as IEEE, please change it in `./oist_thesis.typ`. If you want to cite multiple references side by side, please write them as shown in `@Heald2021-el @Wen2022-ga @Takada2023-du`. result will be a single reference (Shimizu et al., 2023), multile references will be shown as (Heald et al., 2021; Takada et al., 2023; Wen & Imamizu, 2022).

```
@article{Lee98,
  Author = {Lee, Hwang and Scully, Marlan},
  Issn = {0015-9018},
  Issue = {4},
  Journal = {Found. Phys.},
  Pages = {585-600},
  Publisher = {Springer Netherlands},
  Title = {The Physics of EIT and LWI in V-Type Configurations},
  Url = {http://dx.doi.org/10.1023/A:1018709621908},
  Volume = {28},
  Year = {1998}}
```

There are various citation styles, such as APA, Chicago, and IEEE. This template uses APA. If you want to change the citation style, change the `set bibliography(style: "apa", title: "Bibliography")` style parameter in `./oist_thesis.typ`.

3.3.7 Appendices

Structures the style for the appendices and builds them. The appendices are numbered with letters but are structured like regular chapters.

Wrap the body of every appendix file in `#appendix("<letter>")[...]`. The letter drives all the numbering inside the block: sections become `A.1`, `A.1.1`, and figures and tables become `Figure A.1`, `Table A.1`. Every counter restarts for each appendix, so Appendix B starts again at `B.1`. All figures and tables of an appendix must sit inside the block for this to apply.

```
#import "../oist_thesis.typ": appendix, custom_figure

#appendix("A")[
#heading(level: 1, numbering: none, "Appendix A: First Appendix")
<appendix-a>

== A Section Inside an Appendix
...
]
```

3.3.8 Published Articles

This last section add the PDF files of your previously published articles (or about to be published) to the thesis. You should only include PDF files provided by the publishing journal (**including preprint**). This is strictly for the examiners' convenience in the temporary bound thesis, as for copyright reasons these files may not be published in the final version of the thesis.

Use `#embed_pdf("path/to/article.pdf", 15)` to drop all 15 pages of an article into the thesis full-bleed, with no margins or header. Precede it with `#published_article_cover(...)` to generate a cover page carrying the article number, its title, the full citation rendered from `bibliography.bib`, a pointer to the corresponding chapter, and the publisher's copyright notice.

```
#published_article_cover(
  number: "1",
  title: "Title of the published article",
  bib_key: <Lee98>,
  related_chapter: <ch-2>,
  license: "Reproduced with permission from ...",
)
#embed_pdf("published_pdf/study1.pdf", 15)
```

PDFs are large, so keep them out of git (the shipped `.gitignore` already excludes `*.pdf`) and guard the calls with a build flag in `main.typ` so the thesis still compiles on a machine that does not have them:

```
#let embed_published_pdfs = false
...
#if embed_published_pdfs {
  embed_pdf("published_pdf/study1.pdf", 15)
}
```

3.3.9 Differences between draft and submission modes

The `thesis(...mode)` property accepts either `draft` or `submission`. The `draft` mode is used for review, while the `submission` mode is used for the final version of the thesis.

The main difference is the spacing between lines. Therefore, it is likely that the appearance will differ from draft mode, so adjustments may be necessary before submission.

> ***Spacing*** : Spacing should be 1.5 line spaced for examination and single spacing for final submission.
>
> — OIST, Graduate School Policies[3]

[3] https://www.oist.jp/education/policies-regulations/gs-policies

Chapter 4

Figures, tables and images

Here, we describe figures and tables with how to refer.

4.1 Figures

Figures are defined by the `#figure(...)` function. The caption is written in the `caption:` parameter, and the figure itself is written in the first parameter. The caption should be short for the List of Figures, but you can write a full caption in the `caption:` parameter. Refer to figure like this: `@fig-example`. It will be shown as Figure 4.1. If you want to display a shorter form, you can change the style by writing it as in `@fig-example[Fig.]`. It will be shown as Fig. 4.1.

```
#figure(
  image("../images/chap3/emblem.jpg", width: 50%),
  caption: [Full caption with all the details here.font size has to
be 10pt, and sentences has to be left side.],
) <fig-example>
```



Figure 4.1: Full caption with all the details here.font size has to be 10pt, and sentences has to be left side.

The style of figures is defined based on GS Policies.

4.1.1 Short titles for LoF/LoT

If you want to show a short headline (summary) in the List of Figures/Table, use `#custom_figure(...)` and pass a `short:` argument in addition to the full `caption:`. In the body, the caption will appear as "Figure: Short Title Full caption …", while the lists will display only the "Short Title".

```
#custom_figure(
  image("../images/chap3/emblem.jpg", width: 50%),
  short: [Logo, short caption for LoF],
  caption: [Full caption with all the details here. Font size has to
be 10pt, and sentences have to be left-aligned.],
) <fig-example-short>
```



Figure 4.2: Logo, short caption for LoF. Full caption with all the details here. Font size has to be 10pt, and sentences have to be left-aligned.

> ***Spacing*** *: Figures should appear as close as possible to the first mention of them in the text. All figures must be referred to in the text by either a parenthetical mark-up (Figure 1.2), or phrasing such as "Sequencing data, shown in Figure 1.2, shows that…". A parenthetical mention, but not an in-text mention, may be abbreviated as (Fig. 1.2). The number of the chapter should be part of the figure number.*
> *Figures must be accompanied by a caption that describes the material cleanly and succinctly. Figure captions may start with a brief title in bold, which can then be referenced in the list of figures.*
> *As a general rule, figures should not have captions that run across pages. If a figure and its caption will be larger than one page, rewriting should be considered, or a reorganization of the figure. If this cannot be avoided, the figure caption should continue on the immediate next page, with a reference comment at the start of the text to the fact that it is a continuation. No other main body text should then appear on that page.*
>
> — OIST, Graduate School Policies[4]

4.2 Tables

The table can be created as follows.

[4] https://www.oist.jp/education/policies-regulations/gs-policies

```
#figure(
  table(
    columns: (auto, auto),
    align: (center, center),
    [Parameter], [Value],
    line(length: 100%), line(length: 100%),
    $Delta$, "0, 150",
    $alpha$, "85",
    $epsilon$, "6",
    $kappa$, "6.8",
    $gamma$, "0.2",
  ),
  caption: [Short heading for the List of Tables.],
) <tab-values>
```

Table 4.1: tables Short heading for the List of Tables.

Parameter	Value
Δ	0, 150
α	85
ε	6
κ	6.8
γ	0.2

You can also supply short titles for tables using `custom_figure`:

```
#custom_figure(
  table(
    columns: (auto, auto),
    align: (center, center),
    [Parameter], [Value],
    line(length: 100%), line(length: 100%),
    $Delta$, "0, 150",
    $alpha$, "85",
    $epsilon$, "6",
    $kappa$, "6.8",
    $gamma$, "0.2",
  ),
  short: [Parameter Summary],
  caption: [Full table caption with details.],
) <tab-values-short>
```

Refer to tables like this: `@tab-values`, which will be shown as Table 4.1. Shorten form can be written as `@tab-values[Tab.]`, which will be shown as Tab. 4.1.

> ***Tables**: All tables should be referred to in the text by number (for example, "Table 3.1 describes all particles found in…"). Tables may be printed in landscape rather than portrait orientation but must be printed on a separate page with continuing and sequential pagination. Tables may extend for more than one page but should then have the table header row repeated on each page. Arial, Helvetica or Calibri, 10-point font, should be used for tables. Tables should have a heading and may have a caption. The number of the chapter should be part of the table number.*
>
> — OIST, Graduate School Policies[5]

Additional images are attached below (Fig. 4.3, Fig. 4.4).

The first figure shows a Ph.D. candidate writing a thesis by hand. To avoid suspicion of using GPT, this university seems to have reverted to handwritten submissions.



Figure 4.3: A Ph.D. candidate writing his dissertation. Holding a pen, but unable to start the first sentence. Created by Google Nano Banana.

The second figure illustrates a paradigm shift in thesis writing. It evolves from handwriting to Virtual Reality (VR).

[5] https://www.oist.jp/education/policies-regulations/gs-policies

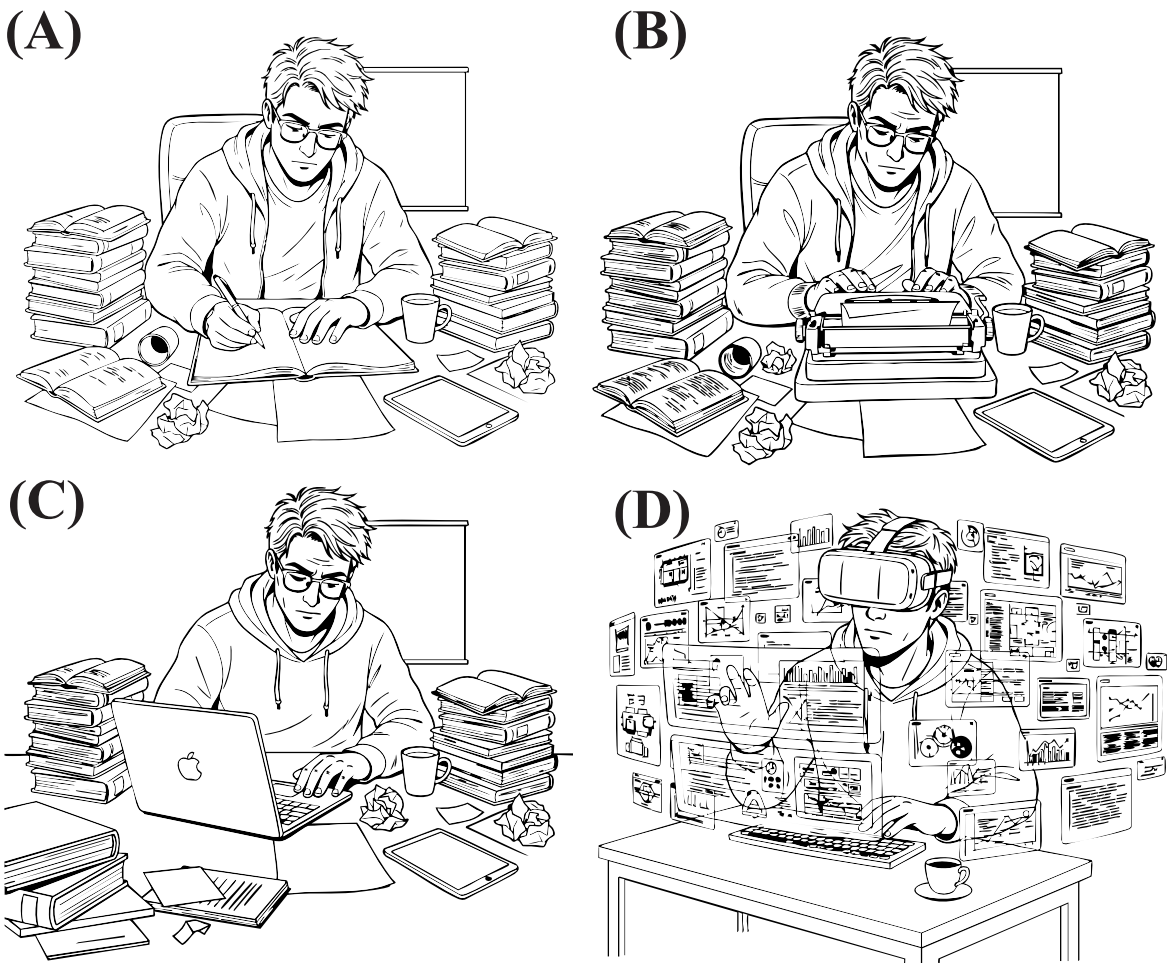


Figure 4.4: The process by which a Ph.D. candidate's tools for writing their dissertation change. (A) Handwritten, (B) Typewriter, (C) Laptop, (D) VR space. The only thing GPT seems unlikely to interfere with is the typewriter. On the other hand, if our arms were driven by Electrical Muscle Stimulation, that might be a different story.

These figures are examples of how to present data in a clear and concise manner.

Chapter 5

Conclusion

This is the conclusion.

Bibliography

Heald, J. B., Lengyel, M., & Wolpert, D. M. (2021). Contextual inference underlies the learning of sensorimotor repertoires. *Nature*, *600*(7889), 489–493.

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Takada, K., Kumasaki, N., Froese, T., Shibata, K., Nishida, J., & Kasahara, S. (2023). ShadowClones: an Interface to Maintain a Multiple Sense of Body-space Coordination in Multiple Visual Perspectives. *Proceedings of the Augmented Humans International Conference 2023, Ahs '23*, 171–178.

Wen, W., & Imamizu, H. (2022). The sense of agency in perception, behaviour and human–machine interactions. *Nature Reviews Psychology*, *1*(4), 211–222.

Appendices

Appendix	Title	Page
A	First Appendix	18
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Appendix A: First Appendix

This is the first appendix. Unlike a numbered chapter, its level-1 heading carries `numbering: none`, so no “Chapter N” prefix is printed.

A.1 A Section Inside an Appendix

Sections inside an appendix are numbered with the appendix letter instead of the chapter number — this one is Section A.1, and the next would be A.2.

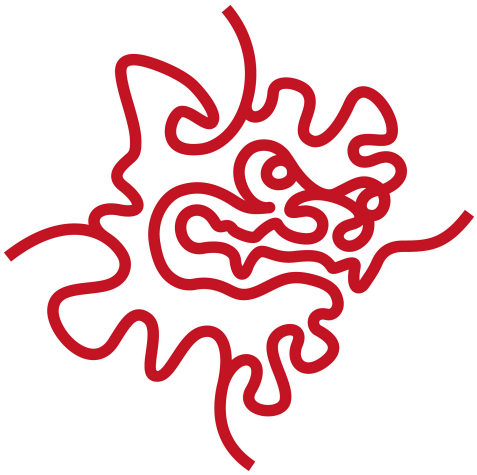


Figure A.1: Appendix figure. Figures inside an appendix are numbered with the appendix letter, so this is Figure A.1 rather than a continuation of the chapter numbering.

Cross-references work as usual: see Figure A.1.

Appendix B: Second Appendix

The second appendix restarts every counter, which is why the section below is B.1 and the table below is Table B.1 — neither continues from Appendix A.

B.1 Another Section

Table B.1: Appendix table. Tables inside an appendix are numbered with the appendix letter, and the counter restarts for every appendix.

Condition	Mean	SD
Baseline	0.42	0.11
Treatment	0.68	0.09

See Table B.1 for an example.